

SMART WORKFORCE AND OPERATIONS MANAGEMENT SYSTEM FOR BUS DEPOTS

Problem Statement ID: skh050 | Theme: Transportation | PS Category: Software

1. Project Overview

1.1 Problem Statement

Bus depots rely heavily on manual processes for staff scheduling, attendance, duty allocation and leave management. These processes can create operational inefficiencies, scheduling conflicts, administrative workload and reduced service reliability.

1.2 Proposed Solution

A digital workforce management platform that centralizes staff, attendance, duty allocation, leave management, scheduling and workforce monitoring. Employees use a mobile interface while authorized depot personnel use management dashboards.

1.3 Core Objectives

- Automate workforce planning and reduce manual administrative effort.
- Provide transparent attendance and leave workflows.
- Improve staff and duty allocation using availability and workload information.
- Provide centralized workforce information, monitoring and reports.
- Support better operational planning and service reliability.

1.4 70% Core Problem + 30% Innovation

Core problem (70%)	Innovation (30%)
Scheduling, attendance, duty allocation, leave, performance and reports	AI assistance, bus fault/emergency hub, geo-tagged cleanliness monitoring.

1.5 One-Sentence Explanation

Our system helps bus depots manage staff attendance, leave and duty scheduling digitally, while smart assistance and operational alerts reduce manual confusion and improve workforce utilization.

2. System Architecture & Technology Stack

2.1 High-Level Architecture

Layer	Function
Users	Admin/depot manager, drivers, conductors and staff
Interface	Employee mobile interface and management dashboard
Application	Scheduling, attendance, leave, notifications and reporting
Processing	Workforce rules, duty allocation, AI assistance and operational workflows
Data	Centralized workforce and operational records
Output	Dashboards, alerts, reports and status updates

2.2 Architecture Flow

USER → MOBILE / DASHBOARD → APPLICATION MODULES → PROCESSING & WORKFLOW LOGIC → DATABASE → DASHBOARD / NOTIFICATION / REPORT

2.3 Key Modules

Module	Purpose
Workforce Scheduling	Organize staff duties using availability and workload.
Attendance Management	Record and monitor staff attendance.
Leave Management	Submit, process and track leave requests.
Duty Allocation	Manage assigned duties.
Performance Dashboard	Provide workforce insights.
Reports & Analytics	Summarize workforce and operational information.
AI Assistance	Guide users through routine workforce tasks.
Fault & Emergency Hub	Route bus fault/repair reports and critical alerts.
Cleanliness Monitoring	Capture daily geo-tagged evidence.

2.4 Technology Stack

The Round 1 PPT identifies Flutter, Node.js, React, HTML/CSS/JavaScript, Bootstrap and MongoDB. The final document must be corrected to match the technologies actually used in the Round 2 prototype; remove any unused technology.

2.5 System Requirements

- Mobile/desktop device with supported application or browser.
- Backend/API runtime and database environment.
- Network connectivity for synchronized data and notifications.
- Role-based access for authorized users.

3. System Design

3.1 User Roles & Access Control

Role	Primary Access
Admin / Depot Manager	Workforce overview, scheduling, attendance, leave, reports and alerts.
Driver / Conductor	Duty, attendance, leave status and relevant alerts.
Employee / Staff	Attendance, leave, duty information and AI assistance.
Mechanic / Maintenance	Bus fault and repair reports.
Cleaner	Daily cleanliness evidence upload where implemented.

3.2 Core Data Entities

Entity	Key Information
User / Staff	Staff ID, role, contact, availability
Attendance	Staff ID, date/time, status, location where applicable
Duty	Staff ID, duty/shift, date/time, status
Leave	Staff ID, type, dates, reason, approval status
Bus / Fault	Bus ID, fault type, details, status, maintenance action
Cleanliness Evidence	Reference, image, timestamp, GPS/location where implemented

3.3 Data Flow

STAFF / PASSENGER / MAINTENANCE INPUT → VALIDATION → APPLICATION WORKFLOW → CENTRAL RECORD → PROCESSING / APPROVAL → DASHBOARD / NOTIFICATION / REPORT

3.4 Interface / API Considerations

- Authentication verifies user role before protected operations.
- Workforce interfaces exchange attendance, duty and leave information with the application layer.
- Notification interfaces deliver duty, leave and emergency updates.
- Fault interfaces create records monitored by authorized personnel.

3.5 User-Adaptive Design

The system is designed around existing user behaviour. Frequently required actions remain accessible, unnecessary navigation is reduced, and additional features are kept focused so the workflow remains simple and adaptable.

4. Technical Workflow & Methodology

4.1 End-to-End Workforce Workflow

LOGIN → ROLE IDENTIFICATION → ATTENDANCE / DUTY / LEAVE INPUT → VALIDATION → CENTRAL DATA STORAGE → SCHEDULING / APPROVAL → DASHBOARD & NOTIFICATIONS → MONITORING

4.2 Attendance Workflow

Step	Technical Action
1	User authenticates using assigned role.
2	Attendance action is submitted.
3	Timestamp and location are captured where implemented.
4	Record is stored against staff profile.
5	Authorized personnel view attendance status.

4.3 Leave Workflow

Employee selects leave → enters details → request stored → authorized person reviews → status updated → employee receives status/notification.

4.4 Scheduling Workflow

Workforce data → availability/workload checks → duty assignment → conflict check → schedule update → staff notification → monitoring.

4.5 AI Assistance — Innovation

User query/request → intent identification → relevant workforce information or guided action → response/instruction → user completes the task. The AI component should be described only to the level actually implemented.

4.6 Bus Fault & Emergency Hub — Innovation

Fault/fire/repair issue → bus/issue details recorded → relevant maintenance personnel notified → status tracked. For critical incidents, an authorized user can broadcast an emergency alert to relevant drivers and conductors.

4.7 Cleanliness — Innovation

Cleaner uploads daily evidence → GPS/location and timestamp where implemented → manager reviews.

5. Implementation

5.1 Module-Wise Implementation

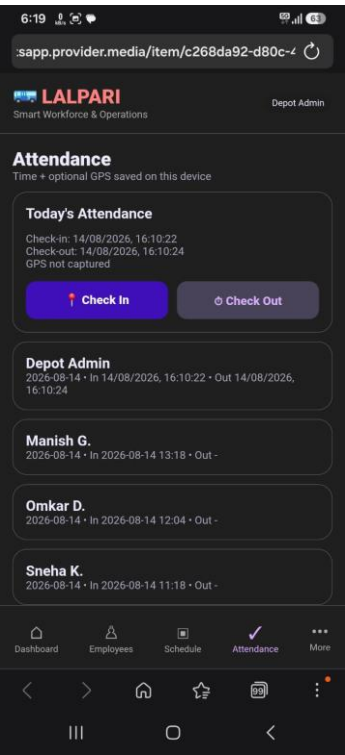
Module	Implementation Focus
Authentication & Roles	Separate access paths for management and workforce users.
Attendance	Digital attendance record and monitoring.
Leave	Request, review/approval and status tracking.
Scheduling	Duty assignment using availability/workload.
Dashboard	Central workforce and operational view.
Notifications	Duty, leave and critical operational notifications.
AI Assistant	Routine task guidance and workforce assistance.
Fault Hub	Bus fault, breakdown, fire and repair reporting.
Cleanliness	Daily photo evidence with GPS/location where implemented.

5.2 Key Technical Components

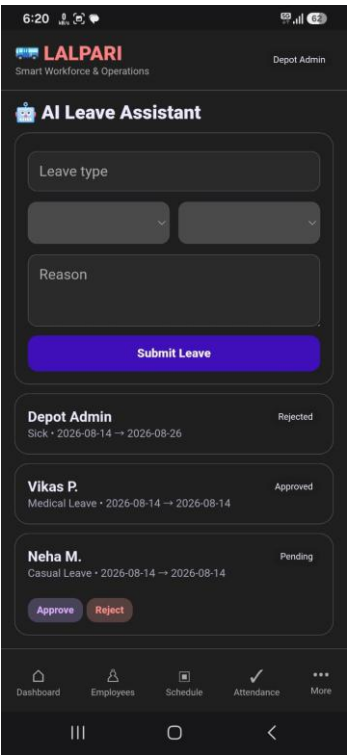
- Role-based interfaces for operational responsibilities.
- Centralized records for staff, attendance, duties, leave and reports.
- Workflow-based approvals and status tracking.
- Dashboard views for monitoring and decision support.
- Notification mechanism for routine and emergency communication.

5.3 Prototype Evidence

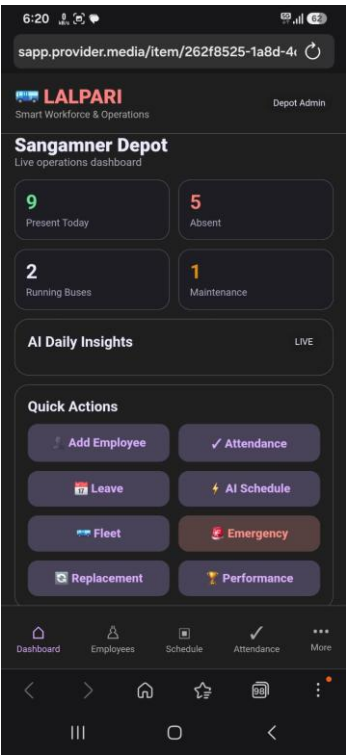
Insert only screenshots corresponding to functions actually demonstrated in the Round 2 prototype.



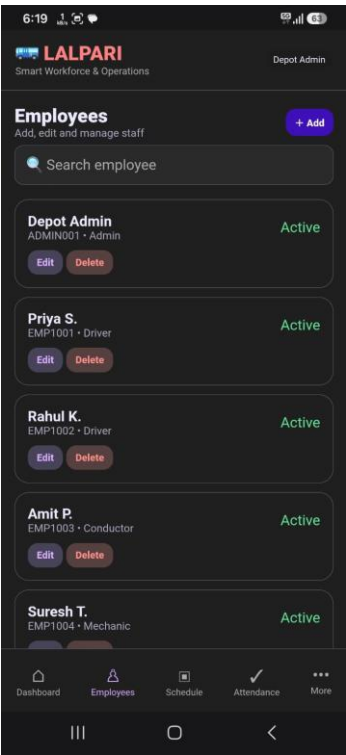
Employee Workflow



Attendance



Leave



Admin Dashboard

5.4 Implementation Integrity

Only developed or demonstrable functions should be labelled as implemented. Features still under development must be identified as prototype/future implementation.

6. Testing & Validation

6.1 Testing Approach

Testing is performed as scenario-based functional validation. Each major workflow is executed using representative staff, duty, attendance, leave and operational scenarios.

6.2 Functional Test Cases

Test Case	Expected Result	Status
Login / role access	Correct role interface displayed	Update after test
Attendance submission	Attendance record created/updated	Update after test
Leave request	Request stored and status visible	Update after test
Duty allocation	Assigned duty displayed	Update after test
Schedule conflict	Conflict identified/reduced	Update after test
Dashboard update	Relevant workforce information displayed	Update after test
Fault report	Issue reaches relevant workflow	Update after test
Emergency alert	Relevant users receive/display alert	Update after test
Cleanliness upload	Evidence stored with location where implemented	Update after test

6.3 Validation Method

- Validate each workflow from the user's perspective.
- Verify submitted data appears in the correct dashboard/status view.
- Check role access and relevant functions.
- Check notifications and status changes.

6.4 Evaluation Metrics

Metric	Evaluation
Workflow completion	Percentage of defined scenarios completed successfully.
Scheduling conflicts	Conflicts before vs. after scheduling logic.
Administrative effort	Manual steps required for a common task.
Notification success	Percentage of tested alerts delivered/displayed correctly.
Usability observation	Time/steps required for common tasks.

7. Results, Limitations & Future Scope

7.1 Prototype Results / Outputs

The prototype demonstrates the transition from manual workforce processes to a centralized digital workflow. The final results must report only the functions actually demonstrated in the prototype and video.

Area	Demonstrable Output
Workforce	Centralized staff information and duty visibility.
Attendance	Digital attendance workflow and monitoring.
Leave	Digital request and approval/status workflow.
Scheduling	Structured duty allocation and conflict-aware planning.
Operations	Dashboard monitoring and notifications.
Innovation	AI assistance, fault/emergency workflow and cleanliness were implemented.

7.2 Impact

- Reduces repetitive manual workforce administration.
- Improves transparency of attendance, leave and duty information.
- Supports better workforce utilization and operational planning.
- Improves communication during operational issues and emergencies.

7.3 Technical Limitations

- Prototype validation uses limited/sample data compared with live deployment.
- Full deployment requires integration with existing depot and transport systems.
- AI assistance and scheduling logic require further validation with real operational data.
- GPS, notification and image workflows depend on device/network permissions and connectivity.

7.4 Future Technical Improvements

- Integrate with live depot workforce and bus-operation databases.
- Improve scheduling optimization using larger historical workforce datasets.
- Add advanced analytics for staffing and operational trends.
- Add validated automated image-based cleanliness analysis when sufficient data is available.
- Scale role-based mobile workflows and notifications.

7.5 Scalability and Feasibility

The modular architecture allows workforce, operational and passenger-facing functions to be expanded independently. Phased deployment can begin with scheduling, attendance and leave management and then add validated operational innovations.

8. References

Retain only references actually consulted for design, implementation or technical decisions.

- Ministry of Road Transport & Highways (MoRTH) — transport-sector reference material.
- National Informatics Centre (NIC) — digital/transport technology resources.
- Google Maps Platform Documentation — location/GPS services, if used.
- Firebase Documentation — authentication/database/notification services, if used.
- Flutter Documentation — mobile application development, if used.
- AWS Cloud Documentation — cloud infrastructure/services, if used.
- IBM AI Documentation — AI concepts/services, if used.

8.1 Final Submission Checklist

Item	Final Check
Documentation	Exactly 8 pages; purely technical.
Architecture / Design	Diagram matches actual system.
Implementation	Only developed/demonstrated modules claimed.
Testing	Actual test results filled in.
Screenshots	Relevant, readable prototype evidence inserted.
YouTube Demo	Actual project/prototype flow demonstrated.
GitHub	Public repository link added if available.
Links	All submitted links tested and accessible.

8.2 Final Technical Message

The project first solves the defined bus-depot workforce-management problem and then extends the platform with four targeted innovations: AI assistance, a bus fault/emergency hub, geo-tagged cleanliness monitoring. The design prioritizes simple, user-adaptive workflows and avoids unnecessary features.

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